

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

$$4x - 3y = 5$$
$$x = 8$$

What is the solution (x, y) to the given system of equations?

Answer

- A. $(8, 9)$
- B. $(8, -24)$
- C. $(8, -9)$
- D. $(8, 24)$

Correct Answer: A

Rationale

Choice A is correct. The second equation in the given system is $x = 8$. Substituting **8** for x in the first equation in the given system yields $4(8) - 3y = 5$, or $32 - 3y = 5$. Subtracting **32** from both sides of this equation yields $-3y = -27$. Dividing both sides of this equation by -3 yields $y = 9$. Therefore, the solution (x, y) to the given system of equations is **$(8, 9)$** .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.